

Homework 2: Simple Linear Regression

ISYE 4031 - Summer 2026 - Problems Only Version

Submission Instructions

Include your full name, Georgia Tech email, and GTID in your submitted work. You may complete the homework using JupyterLite, a local Python or R environment, spreadsheet software, handwritten work where appropriate, or another approved environment. Your submission should clearly answer each question and show enough reasoning, formulas, code output, or supporting work to be graded.

Required Data Files

- `hw2_prob1_accu_copiers.csv`
- `hw2_prob2_labor_cost.csv`

Problem 1: Accu-Copiers Service Time

Accu-Copiers, Inc., sells and services the Accu-500 copying machine. As part of its standard service contract, the company agrees to perform routine services on this copier. To obtain information about the time it takes to perform routine service, the company has collected data for 11 service calls.

Variables:

- x : number of copiers serviced
- y : number of minutes required

Tasks:

1. Show a scatter plot of x and y .
2. Based on the scatter plot, discuss why the simple linear regression model might be a good fit.
3. Calculate the regression line coefficients using the formulas, including the steps.
4. Fit the regression line with software and provide the plot.
5. Interpret the meaning of the slope β_1 .
6. Interpret the meaning of the y-intercept β_0 .
7. If the company wants to service 10 copiers, what is your expected required time?
8. Find an estimate for the variability of the model.

Problem 1(a) Scatter Plot

Use software or code to make a scatter plot. Label both axes and give the plot a meaningful title.

Problem 1(b) Written Answer

Discuss whether a straight-line relationship is reasonable based on the scatter plot. Comment on direction, strength, and whether there are obvious curvature or outlier concerns.

Problem 1(c) Formula Calculation

Use the formulas

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

where

$$S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}), \quad S_{xx} = \sum_i (x_i - \bar{x})^2.$$

Show the quantities you calculate and state the fitted line. You may use software or code to check your arithmetic.

Problem 1(d) Software Fit and Regression Plot

Fit the simple linear regression model with software. Report the fitted coefficients and add the regression line to the scatter plot.

Problem 1(e) Slope Interpretation

Interpret $\hat{\beta}_1$ in the context of copier service time. Be specific about the units of x and y .

Problem 1(f) Intercept Interpretation

Interpret $\hat{\beta}_0$ in context. Also comment on whether this interpretation is practically meaningful for this problem.

Problem 1(g) Prediction at $x = 10$

Use your fitted line to estimate the expected service time for 10 copiers.

Problem 1(h) Variability Estimate

Estimate the model variability. In simple linear regression, this is commonly reported as the residual standard error

$$s = \sqrt{\frac{SSE}{n-2}}.$$

Compute it and explain what it measures in context.

Problem 2: Direct Labor Cost and Batch Size

An accountant wishes to predict direct labor cost y on the basis of the batch size x of a product produced in a job shop. Data for 12 runs are provided in `hw2_prob2_labor_cost.csv`.

Variables:

- x : batch size
- y : direct labor cost

Tasks:

1. Using a scatter plot, discuss whether a linear regression model is a good fit. 2. Use software to fit a regression line. 3. Interpret the meaning of the slope β_1 . 4. Interpret the meaning of the y-intercept β_0 . 5. Find an estimate for the variability of the model.

Problem 2(a) Scatter Plot

Create a scatter plot, then write a short discussion of whether a straight-line model is reasonable.

Problem 2(b) Software Regression Fit

Fit a simple linear regression model predicting direct labor cost from batch size. Report the fitted equation and provide a regression plot.

Problem 2(c) Slope Interpretation

Interpret $\hat{\beta}_1$ in the context of labor cost and batch size.

Problem 2(d) Intercept Interpretation

Interpret $\hat{\beta}_0$ in context. Also comment on whether this interpretation is practically meaningful for this problem.

Problem 2(e) Variability Estimate

Compute the residual standard error and explain what it measures in this context.

AI Usage Report

Your submission must include an AI Usage Report following the course AI policy. If you used AI tools, include the tool(s), what you asked them to help with, what output you accepted, changed, or rejected, and how you verified the final work. If prompts or screenshots materially affected your submission, include copied prompt text or screenshots in the report when appropriate. If no AI tools were used, include this statement: "No AI tools were used for this submission. You may include this report at the end of your submission or attach it as a separate file.